TRAUMA · NEET-SS / FET / INI-SS

Polytrauma, open fractures and compartment syndrome

1. Polytrauma: assessment and priorities

- **Trimodal death distribution (classic):** immediate (brain, heart, great vessels), early "**golden hour**" (haemorrhage, airway, tension pneumothorax: preventable), late (sepsis, multi-organ failure).
- **ATLS primary survey: ABCDE** (catastrophic haemorrhage first in military/C-ABCDE): Airway with cervical spine protection, Breathing, Circulation with haemorrhage control, Disability (GCS, pupils), Exposure.
- **Haemorrhagic shock classes (ATLS):**

Class	Blood loss	Heart rate	BP	Mental state	Base deficit
I	Under 15% (under 750 ml)	Normal	Normal	Slightly anxious	0 to -2
II	15-30%	Mildly raised (100-120)	Normal (narrowed pulse pressure)	Anxious	-2 to -6
III	30-40%	120-140	**Falls**	Confused	-6 to -10
IV	Over 40%	Over 140	Low	Lethargic	Below -10

- **Damage control resuscitation: permissive hypotension** (systolic about 80-90 mmHg, except head injury: MAP over 80), **balanced transfusion (packed cells : plasma : platelets about 1:1:1)**, **tranexamic acid (1 g over 10 min then 1 g over 8 h) within 3 hours** (CRASH-2), avoid crystalloid overload; correct the **lethal triad:**

hypothermia, acidosis, coagulopathy (plus hypocalcaemia: "lethal diamond").

- **Injury Severity Score (ISS):** sum of the **squares of the highest AIS in the three most severely injured body regions** (range 1-75; AIS 6 = 75). **Polytrauma: ISS over 15-17** (or the **Berlin definition**: AIS 3 or more in two or more regions + one physiological factor: hypotension, GCS 8 or less, acidosis BE -6 or less, coagulopathy, age 70 or more).

2. Early total care vs damage control orthopaedics

- **Early total care (ETC):** definitive fixation of all long-bone fractures within 24 hours; reduces pulmonary complications and allows mobilisation in **stable** patients.
- **Damage control orthopaedics (DCO):** in unstable patients, the **"second hit"** of major surgery (reaming, blood loss) on top of the **"first hit"** (injury: SIRS) can cause **ARDS and multiple organ failure**. So: **temporary external fixation**, haemorrhage control, resuscitation in ICU; **definitive fixation once physiology recovers (window about days 4/5-10/14; avoid days 2-4** when the inflammatory response peaks).
- **Patient categories (Pape):**

Status	Criteria (examples)	Strategy
Stable	Haemodynamically stable, no ongoing transfusion, lactate under 2.5	ETC
Borderline	ISS over 40 (or over 20 with chest trauma), multiple long-bone + truncal injury, bilateral lung contusion, mean PAP over 24, hypothermia under 35 C	ETC with caution or DCO
Unstable	Persistent instability despite resuscitation, BP under 90, needing over 2-3 units/hour	**DCO**

Status	Criteria (examples)	Strategy
In extremis	Uncontrolled bleeding, lethal triad	**DCO**: life-saving surgery only

- **Early appropriate care (EAC, Vallier):** definitive fixation within 36 hours once **lactate under 4 mmol/L, pH 7.25 or above, base excess -5.5 or above** (resuscitation endpoints), rather than fixed time rules.
- **Fat embolism syndrome:** 24-72 hours after long-bone/pelvic fractures; **Gurd and Wilson criteria** (major: **respiratory insufficiency, cerebral involvement, petechial rash** in axillae/conjunctivae/oral mucosa; minor: tachycardia, fever, retinal changes, jaundice, renal signs, anaemia, thrombocytopenia, raised ESR, fat macroglobulinaemia). Prevention: **early fracture stabilisation**; treatment supportive (oxygen, ventilation).

3. Open fractures

Gustilo-Anderson classification (graded **after debridement**)

Type	Wound	Soft tissue	Contamination / fracture
I	**Under 1 cm**	Minimal	Clean; simple fracture
II	**1-10 cm**	Moderate	Moderate contamination
IIIA	Usually over 10 cm	Extensive	**Adequate soft-tissue coverage of bone** despite injury; includes high-energy/segmental/comminuted and **farmyard or gunshot** injuries regardless of size

Type	Wound	Soft tissue	Contamination / fracture
IIIB	Extensive	**Periosteal stripping, bone exposed: needs flap cover**	Heavy contamination
IIIC	Any	Any	**Arterial injury requiring repair**

- **Infection rates:** I about 0-2%, II about 2-7%, IIIA about 7%, IIIB about 10-50%, IIIC about 25-50%; amputation rates rise to about 40-50% in IIIC tibial fractures.
- **OTA Open Fracture Classification (skin, muscle, arterial, contamination, bone loss)** is more reproducible.

Management (BOAST 4 / BAPRAS-BOA standards; AAOS)

1. **Antibiotics as soon as possible, ideally within 1 hour** of injury (the most important factor in reducing infection). **Cefazolin (first-generation cephalosporin)** for all; add **gram-negative cover (gentamicin)** for type III; **penicillin (anaerobic/clostridial cover)** for farmyard/soil contamination (AAOS 2022 guidance: cefazolin for I/II, ceftriaxone or cefazolin + gentamicin for III). Continue until **24 hours after wound closure** (no longer than 72 hours total).
2. **Tetanus prophylaxis** (toxoid ± immunoglobulin by status).
3. **Photograph** the wound, cover with **saline-soaked dressing**, **realign and splint**, document neurovascular status. Do not irrigate or probe in the emergency department.
4. **Debridement (excision)** by senior surgeons: within **12 hours** for high-energy injuries, **24 hours** for others, **immediately** for gross contamination, vascular compromise or compartment syndrome. Extend the wound along fasciotomy lines, remove all dead tissue (**4**

Cs for muscle: colour, consistency, contractility, capacity to bleed), loose bone fragments without soft-tissue attachment.

5. **Irrigation: low-pressure, normal saline** (**FLOW trial**: **soap was worse than saline**; low pressure equal to high pressure, cheaper). Volume by Gustilo type (3, 6, 9 L traditionally).
6. **Stabilisation: external fixation** or **definitive internal fixation (intramedullary nail)** if the wound is clean and cover is planned. **Unreamed vs reamed nails: no difference (SPRINT)**.
7. **Soft-tissue cover: definitive cover within 72 hours** ("**fix and flap**" by a combined orthoplastic team), ideally at the same time as definitive fixation. **Negative-pressure wound therapy** as a temporary dressing only (**WOLLF trial**: no benefit over standard dressing for infection or function).
8. **Local antibiotic** (beads, coated nails) for type III injuries.

Limb salvage vs amputation

- **Absent plantar sensation is NOT an indication for amputation** (it was once taught as one). In the **LEAP study**, about half of initially insensate feet regained sensation by 2 years, and **salvage and amputation had similar functional outcomes (Sickness Impact Profile) at 2 and 7 years**.
- The decision is individual: **warm ischaemia over about 6 hours**, a non-reconstructable vascular injury, severe soft-tissue and bone loss, a poor host, and the patient's wishes. Secondary amputation remains an option if salvage fails.
- **MESS (Mangled Extremity Severity Score):** skeletal/soft-tissue energy, limb ischaemia (doubled if over 6 hours), shock, age; **7 or more** historically suggested amputation (**poor predictive value; not to be used alone**). Others: LSI, PSI, NISSSA, HFS-97.

Gunshot injuries

- **Low-velocity (handgun)**: often managed like low-grade open fractures (local wound care, oral/short IV antibiotics). **High-velocity**

(rifle) or shotgun at close range: **cavitation**, extensive debridement (treat as type III). **Intra-articular bullets** must be removed (lead synovitis/toxicity, mechanical damage). Bullets in the bowel then through bone: broad-spectrum antibiotics and debridement.

4. Vascular injury

- **Hard signs** (pulsatile bleeding, expanding haematoma, absent pulses, bruit/thrill, ischaemia): **immediate exploration/repair**. **Soft signs:** ABI.
- **Ankle-brachial index (ABPI) under 0.9 = arterial study (CT angiography)**; over 0.9 has a high negative predictive value.
- High-risk injuries: **knee dislocation** (popliteal artery; **ABI for every knee dislocation**), supracondylar humerus, Schatzker IV/VI plateau, distal femur.
- **Sequence:** temporary **vascular shunt** first (restores flow within the golden **6 hours**), then skeletal stabilisation, then definitive vascular repair, **prophylactic fasciotomy** after reperfusion (especially ischaemia over 4-6 hours).

5. Compartment syndrome

Pathophysiology and causes

- Raised pressure within a closed fascial compartment exceeds capillary perfusion pressure: venous congestion, muscle and nerve ischaemia (**muscle necrosis after about 4-6 hours, irreversible by 6-8 hours**; nerve function lost earlier).
- **Causes: tibial shaft fractures (commonest, about 2-10%; young men)**, forearm fractures, crush injury, **reperfusion after vascular repair**, tight casts/dressings, burns (eschar), prolonged lithotomy, drug overdose with limb compression, snake bite, bleeding disorders/anticoagulation, IV infiltration. **Open fractures do not exclude it.**

- Highest-risk groups: **young men** (muscle bulk), tibial and forearm fractures, high-energy injuries, **unconscious/ventilated/regional anaesthesia** patients (signs masked).

Diagnosis (clinical first)

- **Pain out of proportion** to the injury, **pain on passive stretch** of muscles in the compartment (earliest, most sensitive signs), **tense swelling**, then paraesthesia/hypoaesthesia (deep peroneal: first web space in anterior compartment), weakness. **Pulses are usually present** until late (pulselessness is a late sign, or indicates arterial injury).
- **Compartment pressure measurement** (slit catheter/arterial line transducer, within 5 cm of the fracture, deep posterior and anterior compartments of the leg): **delta pressure (diastolic BP minus compartment pressure) under 30 mmHg** = fasciotomy (McQueen). Absolute pressure over about 30-45 mmHg was older criteria. **Continuous monitoring** in obtunded patients. Diastolic pressure under anaesthesia is lower: use pre-operative diastolic values.

Treatment: emergency fasciotomy

- Remove all circumferential dressings and casts to skin, place the limb at heart level (**not elevated**), correct hypotension, oxygen; then **fasciotomy without delay**.
- **Leg: four compartments** (anterior, lateral, superficial posterior, deep posterior):
  - **Two-incision technique: anterolateral incision** (about 2 cm anterior to the fibula; anterior and lateral compartments; **superficial peroneal nerve** at risk) and **posteromedial incision** (about 2 cm posterior to the medial tibial border; superficial and deep posterior compartments; **saphenous nerve and vein**). Release the deep posterior compartment by detaching soleus from the tibia.
  - Single-incision (lateral, perifibular) alternative.

- **Thigh:** three compartments (anterior, posterior, medial); lateral incision releases anterior and posterior.
- **Forearm: volar (Henry-type curvilinear) incision** from medial to the biceps tendon to the carpal tunnel (release **carpal tunnel and lacertus fibrosus**), ± dorsal incision for the mobile wad and dorsal compartment. **Deep flexors (FDP, FPL)** are most affected.
- **Hand:** 10 compartments (4 dorsal interossei, 3 palmar interossei, thenar, hypothenar, adductor pollicis): dorsal incisions over the 2nd and 4th metacarpals, thenar and hypothenar incisions; plus carpal tunnel release.
- **Foot:** 9 compartments (medial, lateral, 4 interossei, 3 central: superficial, adductor, calcaneal): **calcaneal fractures and crush injuries**; dorsal two-incision + medial incision.
- **Wounds left open**; re-look at 48-72 hours; delayed closure or skin graft.
- **Missed/late compartment syndrome (over about 24-48 hours, established necrosis):** fasciotomy is **not** recommended (high infection risk; converts closed necrosis to open infected necrosis); manage late contracture (**Volkmann ischaemic contracture**: flexor slide (**Max Page muscle slide**), tendon transfers, free functioning muscle transfer).
- **Complications:** rhabdomyolysis and **myoglobinuric renal failure**, hyperkalaemia, infection, contractures, sensory loss, chronic pain.

Chronic exertional compartment syndrome

- Athletes (runners); pain during exercise relieved by rest; **anterior and lateral leg compartments**. **Pressure criteria (Pedowitz): pre-exercise 15 mmHg or more, 1 minute after exercise 30 or more, 5 minutes after 20 or more.** Fasciotomy if conservative care fails.

6. Crush syndrome and rhabdomyolysis

- Prolonged compression (earthquakes, entrapment): on release, **hyperkalaemia (cardiac arrest)**, **myoglobinuria (acute kidney injury)**, hypovolaemia, metabolic acidosis, DIC.
- **Treatment: aggressive IV fluids before extrication if possible** (normal saline, aiming for urine output about 200-300 ml/hour), treat hyperkalaemia, consider bicarbonate/mannitol (debated), dialysis.

> Exam pearl: **ISS = sum of squares of the three worst AIS regions (polytrauma over 15-17). DCO for unstable/borderline patients; definitive surgery after day 4-5, avoid days 2-4.** TXA within 3 hours.

> Exam pearl: **Gustilo is graded after debridement; IIIB = flap needed; IIIC = arterial repair. Antibiotics within 1 hour; FLOW: saline, low pressure, no soap; definitive cover within 72 hours. LEAP: plantar insensation is not an indication for amputation.**

> Exam pearl: **Compartment syndrome: pain on passive stretch is earliest; delta pressure (diastolic - compartment) under 30 mmHg = fasciotomy; pulses are usually present.** Leg: two-incision, four-compartment release.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.